

ELECTRONIC PRODUCT DESIGN PROJECT

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- **Design a circuit with input 24 V DC (Assume a 24 battery) and output of +15V and 0.5 A**

For the given design requirements, A buck convertor is ideal for this application, moreover, we will be using a "Jellybean", a Widely Available and Affordable Convertor, for the design. Texas Instruments, MC34063 is ideal for our use case

Design Requirements: - Referencing the Chapter 9.2.3.1 of the Datasheet, we need to define certain design requirements given below

$$V_{in} = 24V$$

$$V_{out} = 15V$$

$$I_{out(Max)} = 0.5A$$

$$f_{min} = 100 \text{ kHz (Maximum Switching Frequency, refer 8.3, taken to reduce Component Values)}$$

$$V_{ripple} = \pm 100mV \text{ (Peak to Peak Value of desired Ripple)}$$

$$V_{sat} = 1V \text{ (Refer 7.6 For Saturation Voltage of Darlington pair)}$$

$$V_f = 0.4V \text{ (Refer Figure 13 of Diode 1N5819)}$$

$$R_1 = 1000 \Omega$$

$$R_2 = 11000\Omega$$

Design Procedure: - Referencing Table 9.2.3.2, we need to calculate various parameters to determine the component Values

$$\frac{t_{on}}{t_{off}} = \frac{V_{out} + V_f}{V_{in(Min)} - V_{Sat} - V_{Out}} = \frac{15 + 0.4}{24 - 1 - 15} = \frac{77}{40} = 1.925$$

$$(t_{on} + t_{off}) = \frac{1}{f} = \frac{1}{100000} = 1 \times 10^{-5} \text{ sec}$$

$$t_{off} = \frac{t_{on} + t_{off}}{\frac{t_{on}}{t_{off}} + 1} = \frac{1 \times 10^{-5}}{1.925 + 1} = \frac{1}{292500} = 3.418803419 \times 10^{-6} \text{ sec}$$

$$t_{on} = (t_{on} + t_{off}) - t_{off} = 1 \times 10^{-5} - \frac{1}{292500} = 6.581196581 \times 10^{-6}$$

$$C_T = 4 \times 10^{-5} t_{on} = 2.6323478632 \times 10^{-10} \text{ F} \approx 270pF$$

$$I_{pk(Switch)} = 2 \times I_{out(Max)} = 2 * 0.5 = 1A$$

$$R_{sc} = \frac{0.3}{I_{pk(Switch)}} = 0.3 \Omega$$

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$$L_{Min} = \left(\frac{V_{in(Min)} - V_{sat} - V_{out}}{I_{pk(switch)}} \right) t_{on_{max}} = \left(\frac{24 - 1 - 15}{1} \right) * 6.581196581 \times 10^{-6} \\ = 5.264957265 \times 10^{-5} \approx 52\mu H \approx 68\mu H \text{ (Next standard value)}$$

$$C_O = \frac{I_{pk(switch)} * (t_{on} + t_{off})}{8V_{ripple(pp)}} = \frac{1 * 1 \times 10^{-5}}{8 * 0.2} = 6.25 \times 10^{-6} \approx 6.8\mu F$$

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1} \right) = 1.25 \left(1 + \frac{11000}{1000} \right) = 15V$$

Final Component Values

Timing Capacitor: - $C_T = 270 \text{ pF}$

Output Capacitor: - $C_O = 6.8 \mu F$

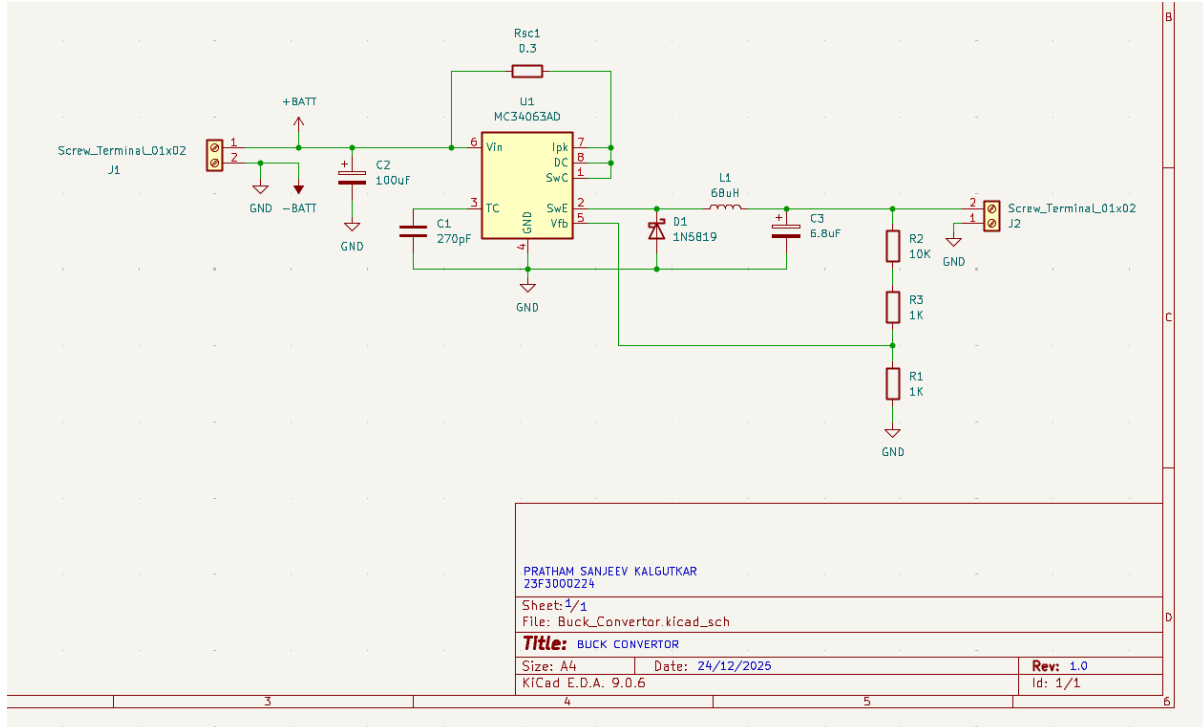
Inductor: - $L = 68 \mu H$

Short Circuit Resistor: - $R_{sc} = 0.3 \Omega$

Feedback Resistors: - $R_1 = 1K\Omega$

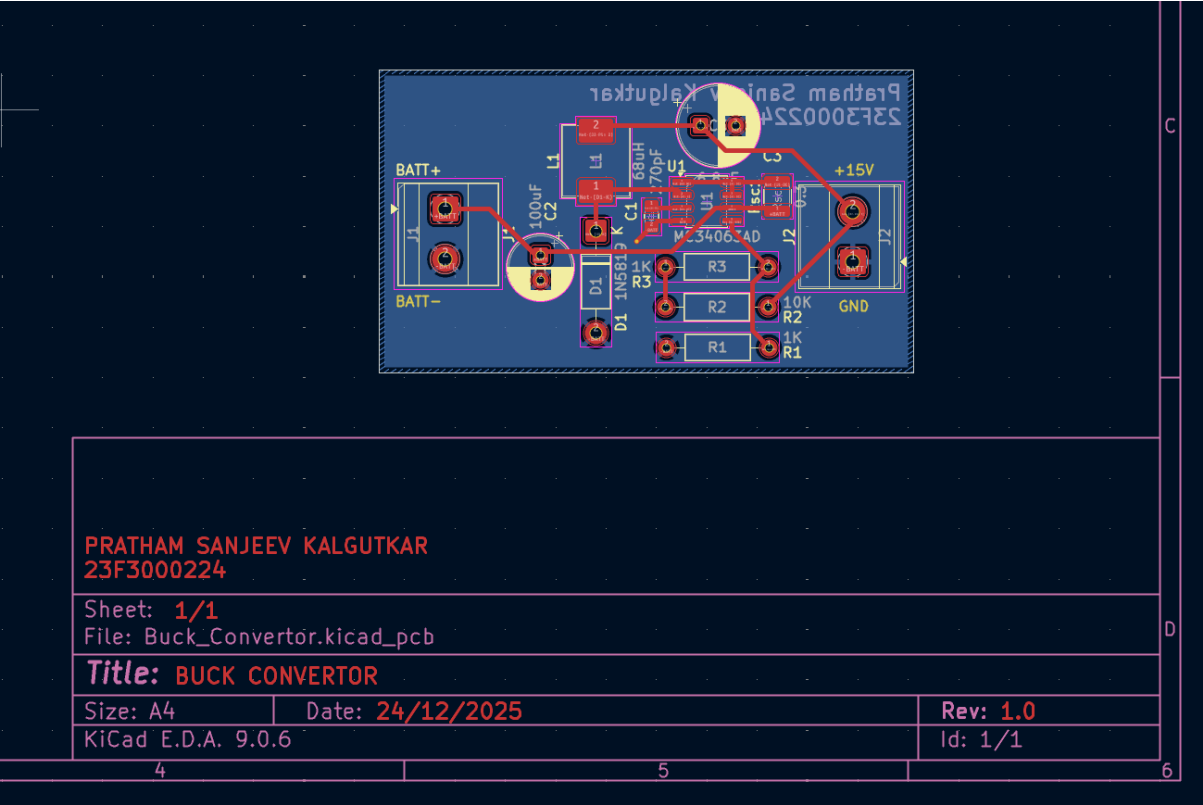
$R_2 = 11K\Omega$

Schematic Diagram: -

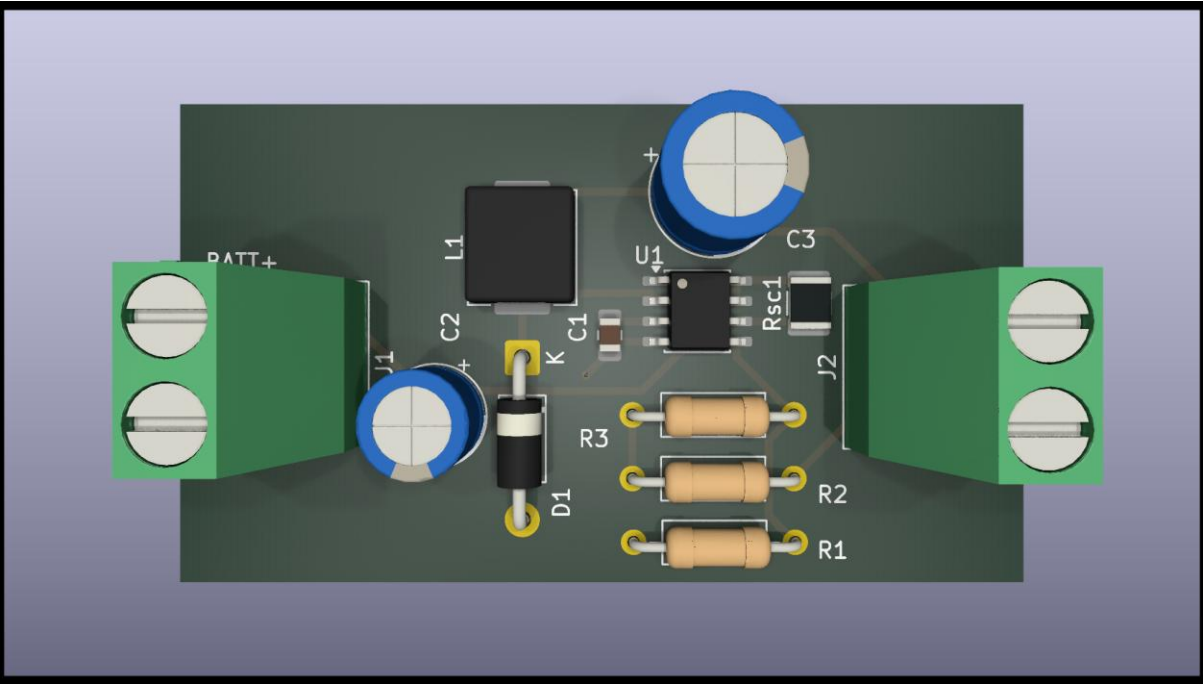


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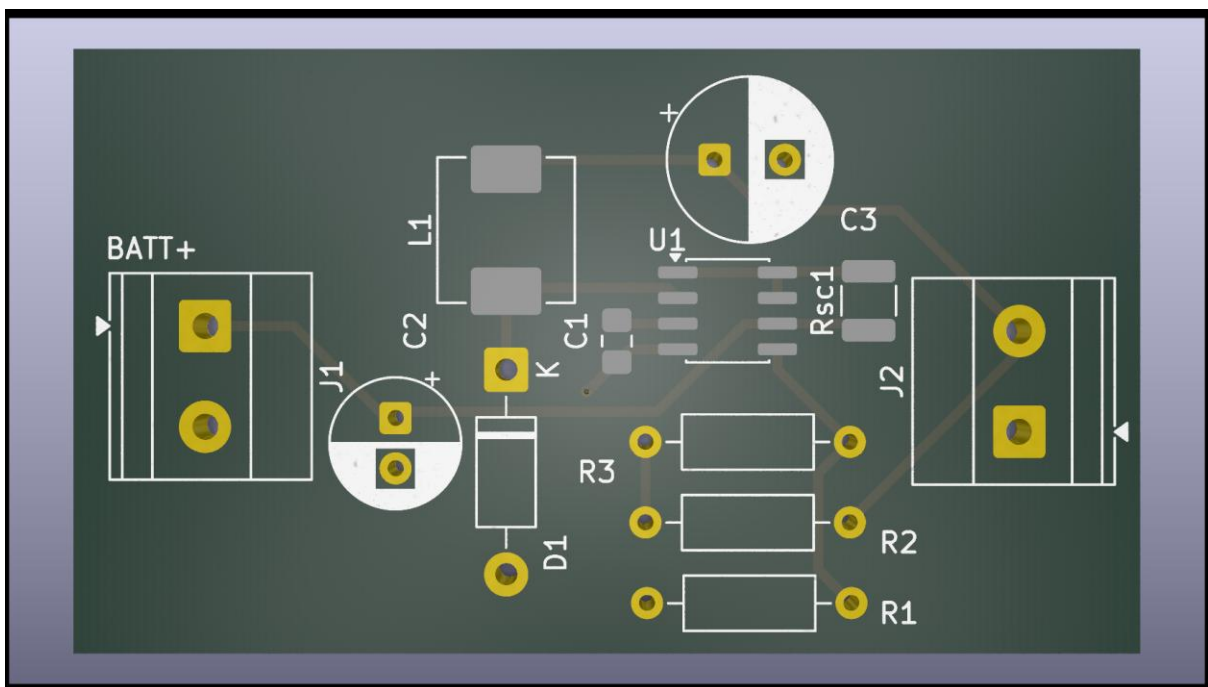
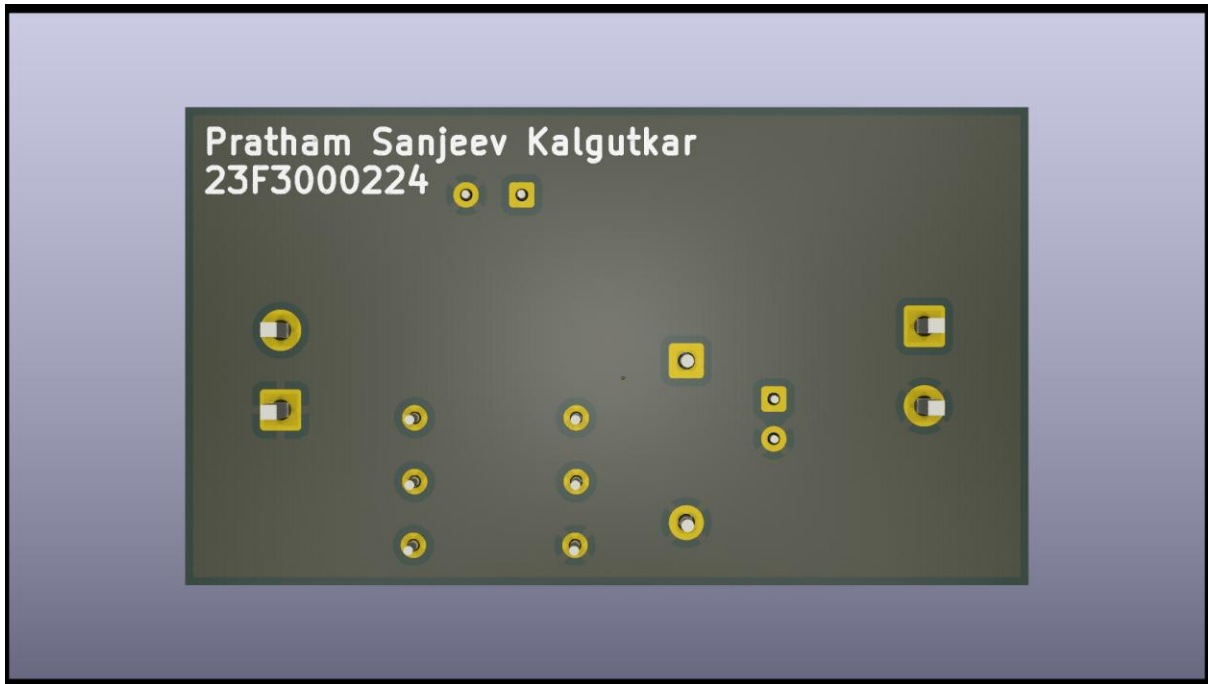
PCB LAYOUT: -



3D View: -



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Estimated Output Calculation: -

$$\text{Output Power} = V_{out} \times I_{Max} = 15 \times 0.5 = 7.5W$$

$$Shunt\ Loss = I^2 R_{sc} = 0.5^2 \times 0.3 = 0.075W$$

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$$\text{Switch Conduction Loss} = I_{out} \times V_{sat} \times \frac{\frac{t_{on}}{t_{off}}}{\frac{t_{on}}{t_{off}} + 1} = 0.5 * 1 * \frac{1.925}{1.925 + 1} = \frac{77}{234} \approx 0.329059W$$

$$\text{Diode Loss} = I_{out} \times V_f \times \left(1 - \frac{\frac{t_{on}}{t_{off}}}{\frac{t_{on}}{t_{off}} + 1}\right) = 0.5 \times 0.4 \times \left(1 - \frac{1.925}{1.925 + 1}\right) = \frac{6}{117} \approx 0.068W$$

$$\text{Inductor Loss} = I^2 \times R_{DC} = 0.5^2 \times 0.213 = 0.05325W$$

$$\text{Quiescent Power Draw} = V_{in} * I_{cc} = 24 * 0.004 = 0.096W$$

$$\text{Switching Loss} = 0.5 \times I_{pk} \times V_{in} \times f \times t_{rise} = 0.5 \times 1 \times 24 \times 100000 \times 75ns = 0.09W$$

$$\text{Total Loss} = 0.075 + \frac{77}{234} + \frac{6}{117} + 0.05325 + 0.096 + 0.9 = 0.4558019603W$$

$$\text{Output Power} = \text{Input Power} + \text{Total Loss} = 7.5 + 0.4580419603 = 7.95580196W$$

$$\text{Estimated Efficiency } \eta \approx \frac{\text{Output Power}}{\text{Input Power}} = \frac{7.5}{7.95580196} = 0.942708232 \approx 94.27\%$$

BILL OF MATERIALS: -

#	Reference	Qty	Value	Datasheet	Product Link	Price
1	Rsc1	1	0.3	~	https://robu.in/product/0-3-ohm-3w-surface-mount-sense-resistor-pack-of-3/#tab-description	12
2	R2	1	10K	~	https://robu.in/product/10k-ohm-0-5w-metal-film-resistor-pack-of-50/	0.98
3	R1, R3	2	1K	~	https://robu.in/product/1k-ohm-0-5w-metal-film-resistor-pack-of-50/	0.98
4	J1, J2	2	Screw_Terminal_01x02	~	https://robu.in/product/2-pin-straight-pcb-screw-terminal-block-connector-3-5mm-pitch/ https://robu.in/product/400ax6r8mefct78x10-8-rubycon-6-8uf-400v-%c2%b120-plugind8xl10-8mm-aluminum-electrolytic-capacitors-leaded-rohs/#tab-description	10
5	C3	1	6.8uF	~	https://robu.in/product/0-3-ohm-3w-surface-mount-sense-resistor-pack-of-3/#tab-description	13
6	C2	1	100uF	~	https://robu.in/product/c0805c271j5gac	50
7	C1	1	270pF	~	tu-kemet-smd-multilayer-ceramic-capacitor-270-pf-50-v-0805-2012-metric-%c2%b1-5-c0g-np0-c-series-kemet/	1.83

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8	D1	1	1N5819	https://www.st.com/resource/en/datasheet/1n5819.pdf	https://robu.in/product/1n5819-1w-diode-pack-of-30/	2.67
9	L1	1	68uH	https://products.sumida.com/products/pdf/CDRH104R_202403.pdf?t=1766595187	https://robu.in/product/cdrh104r-68%ce%bch-power-inductor-pack-of-5/#tab-specification	15
10	U1	1	MC34063AD	http://www.onsemi.com/pub_link/Collateral/MC34063A-D.PDF	https://robu.in/product/mc34063ad-texas-instruments-adjustable-1-25v40v-3v40v-1-5a-soic-8-dc-dc-converters-rohs/	43 149.46

KiCAD Files can be accessed from [here](#)

<https://github.com/PrathamKalgutkar/Buck-Convertor>

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